

IT-SIMPLERA INSTITUTE

Cyber Security Analyst Internship Program



WEEK 01

ASSIGNMENT

Cybersecurity Fundamentals
& Security Concepts

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1. Introduction

Cybersecurity is the practice of protecting systems, networks, and programs from digital attacks. These cyberattacks are usually aimed at accessing, changing, or destroying sensitive information, extorting money from users, or interrupting normal business processes. In today's increasingly connected world, cybersecurity has become one of the most critical areas in information technology.

This report is prepared as part of the **IT-SIMPLERA Cyber Security Analyst Internship Program** (Week 01 Assignment). The objectives include developing a foundational understanding of cybersecurity principles, learning the CIA Triad, researching real-world cyberattacks, exploring common cybersecurity tools, designing a secure home network, and understanding the daily responsibilities of a cybersecurity analyst.

Key Learning Goals: CIA Triad principles, real-world attack analysis, cybersecurity toolsets, network security architecture, and analyst role comprehension.

2. Task 1 – Five Real-World Cyberattacks

The following table documents five significant real-world cyberattacks that have shaped the cybersecurity landscape. Each attack is analyzed by name, year, targeted organization, attack type, description, and impact.

Attack Name	Year	Organization	Type	Description	Impact
WannaCry Ransomware	2017	Global (NHS, FedEx, etc.)	Ransomware	A massive ransomware attack exploiting the EternalBlue vulnerability in Microsoft Windows. The malware encrypted user files and demanded Bitcoin ransom payments of \$300-\$600.	Affected 200,000+ computers across 150 countries. NHS hospitals in the UK were severely disrupted, causing cancelled appointments and diverted ambulances. Estimated damages exceeded \$4 billion.
Equifax Data Breach	2017	Equifax	Data Breach	Attackers exploited an unpatched Apache Struts vulnerability (CVE-2017-5638) in Equifax's web application to gain unauthorized access to sensitive databases over several months.	Personal data of 147 million people compromised, including Social Security numbers, birth dates, and addresses. Equifax paid over \$700 million in settlements.

Attack Name	Year	Organization	Type	Description	Impact
SolarWinds Supply Chain	2020	SolarWinds / US Government	Supply Chain Attack	State-sponsored attackers inserted malicious code (SUNBURST) into SolarWinds' Orion software updates, creating a backdoor into thousands of organizations' networks.	Compromised 18,000+ organizations including US Treasury, Homeland Security, and Fortune 500 companies. Considered one of the most sophisticated espionage operations in history.
Colonial Pipeline	2021	Colonial Pipeline Co.	Ransomware	DarkSide ransomware group attacked Colonial Pipeline's IT systems through a compromised VPN password. The company proactively shut down its OT pipeline systems.	Caused fuel shortages across the US East Coast for nearly a week. Colonial paid \$4.4 million in ransom (partially recovered). Led to an Executive Order on cybersecurity.
Yahoo Data Breach	2013-2014	Yahoo Inc.	Data Breach / State-Sponsored	Two separate breaches: one in 2013 affecting all 3 billion accounts, and one in 2014 affecting 500 million accounts. Attackers used forged cookies and stolen backup keys.	Largest data breach in history. User names, emails, dates of birth, and hashed passwords exposed. Yahoo's sale price to Verizon was reduced by \$350 million as a result.

Table 1: Five significant real-world cyberattacks and their impacts.

3. Task 2 – CIA Triad Violation Analysis

The **CIA Triad** is the foundational model of information security, comprising three core principles:

C	Confidentiality Ensuring that information is accessible only to those authorized to access it. Breaches of confidentiality result in unauthorized disclosure of sensitive data.
I	Integrity Maintaining the accuracy and completeness of data. Integrity violations occur when data is altered, tampered with, or destroyed by unauthorized parties.
A	Availability Ensuring that information and resources are accessible to authorized users when needed. Attacks on availability disrupt access to critical systems and services.

CIA Triad Violations per Attack

Attack	CIA Violated	Explanation
WannaCry Ransomware	Availability & Confidentiality	The ransomware encrypted files, making them unavailable (Availability). Attackers could also access and exfiltrate data before encryption (Confidentiality).
Equifax Data Breach	Confidentiality	Personal data of 147 million individuals was exposed to unauthorized parties, directly violating confidentiality of sensitive records.
SolarWinds Attack	Confidentiality & Integrity	Attackers gained covert access to confidential government data (Confidentiality). The insertion of malicious code into legitimate software updates compromised system integrity (Integrity).
Colonial Pipeline	Availability	Ransomware forced shutdown of the pipeline's operational technology systems, disrupting fuel supply across the US East Coast — a direct availability violation.
Yahoo Data Breach	Confidentiality & Integrity	Billions of user credentials were exposed (Confidentiality). Forged cookies allowed attackers to tamper with authentication mechanisms (Integrity).

Table 2: CIA Triad violation analysis for each cyberattack.

4. Task 3 – Cybersecurity Tools Research

Cybersecurity professionals rely on a diverse toolkit to monitor, detect, analyze, and respond to security threats. The following three tools are widely used across the industry.

W	Wireshark Network Protocol Analyzer
Function: Wireshark captures and interactively analyzes network traffic in real time. It allows security analysts to inspect packets at a granular level, decode hundreds of protocols, and identify anomalies, malicious payloads, or unauthorized communications on a network.	
Use Case: A security analyst uses Wireshark to investigate a suspected data exfiltration incident by capturing network traffic and filtering for unusual outbound connections to unknown IP addresses. It is also used for troubleshooting network performance issues and verifying firewall rules.	
N	Nmap (Network Mapper) Network Discovery & Security Auditing
Function: Nmap scans networks to discover hosts, open ports, running services, and operating system versions. It supports various scan types (SYN, TCP connect, UDP, stealth) and can detect vulnerabilities using its scripting engine (NSE).	
Use Case: Before a penetration test, an analyst uses Nmap to scan the target network, identifying all live hosts, open ports, and services. This reconnaissance helps map the attack surface and prioritize which systems to test for vulnerabilities.	
S	Splunk SIEM – Security Information & Event Management
Function: Splunk collects, indexes, and correlates machine-generated data (logs, events, metrics) from across an organization's IT infrastructure. It provides real-time monitoring, alerting, dashboards, and advanced search capabilities for security operations.	
Use Case: A SOC (Security Operations Center) team uses Splunk to aggregate logs from firewalls, IDS/IPS, servers, and endpoints. Automated alerts are configured to detect brute-force login attempts, privilege escalation, or lateral movement patterns indicative of a breach.	

Table 3: Three essential cybersecurity tools and their applications.

5. Task 4 – Secure Home Network Design

A secure home network is the first line of defense for personal cybersecurity. The following diagram illustrates a well-designed home network architecture incorporating multiple layers of security: a firewall-enabled router, network segmentation, antivirus on endpoints, optional VPN, and strong Wi-Fi encryption (WPA3).

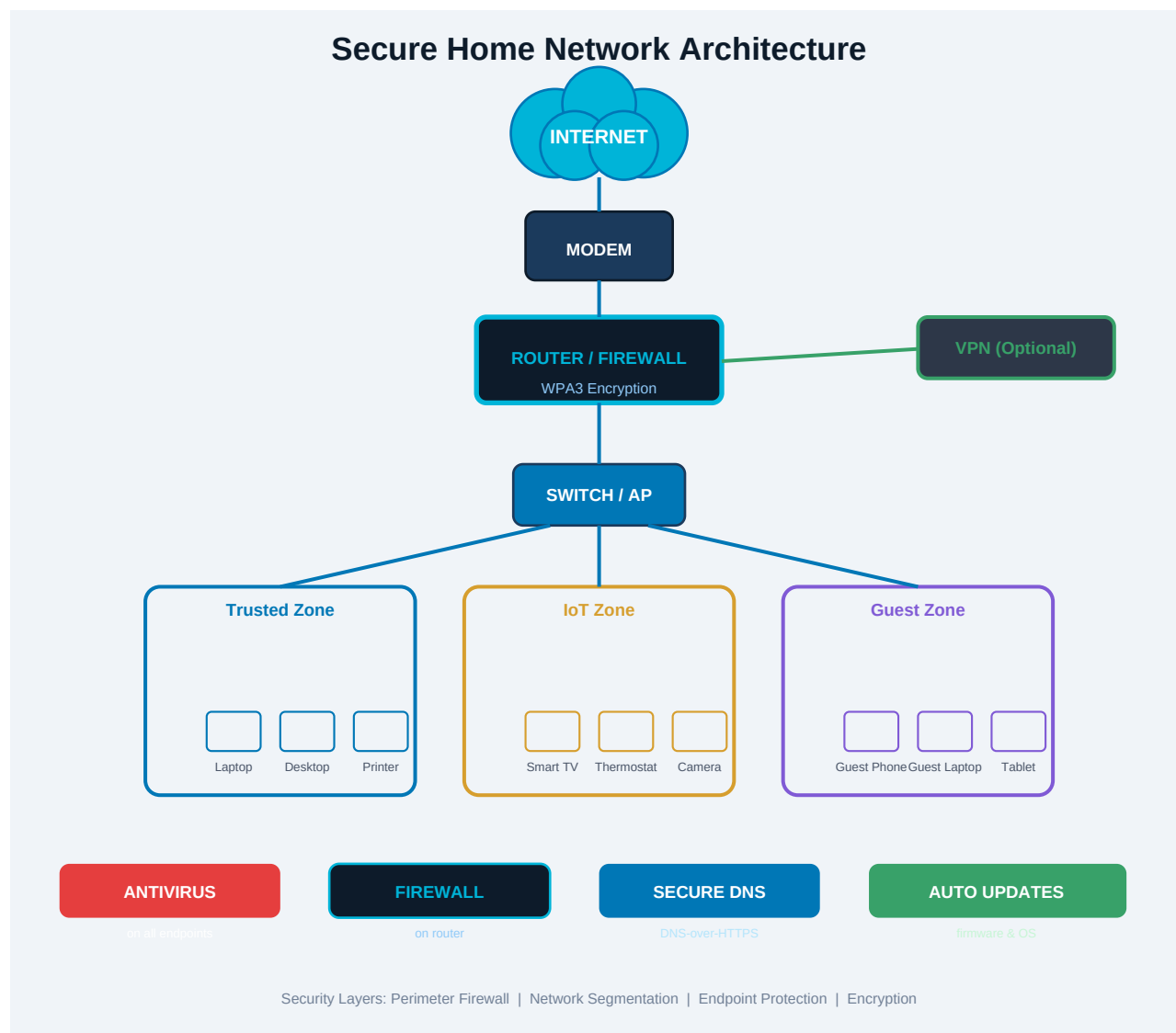


Figure 1: Secure Home Network Architecture Diagram

Security Features Implemented

- **Router Firewall (NAT + SPI):** Blocks unsolicited inbound traffic and inspects packets for known attack patterns.
- **WPA3 Wi-Fi Encryption:** Provides the strongest consumer-grade wireless encryption with forward secrecy.
- **Network Segmentation:** Isolates IoT devices and guest devices from trusted personal devices, limiting lateral movement.

- **Antivirus/EDR on Endpoints:** Detects and removes malware, ransomware, and suspicious processes on each device.
- **VPN (Optional):** Encrypts all internet traffic, protecting privacy especially on untrusted networks.
- **Secure DNS (DoH/DoT):** Prevents DNS hijacking and snooping by encrypting DNS queries.
- **Automatic Firmware Updates:** Ensures router and devices have the latest security patches.

6. Task 5 – What Does a Cybersecurity Analyst Do Every Day?

A cybersecurity analyst serves as the frontline defender of an organization's digital assets. Their daily responsibilities are diverse, dynamic, and critical to maintaining an organization's security posture. The role demands a blend of technical expertise, analytical thinking, and effective communication.

The day typically begins with reviewing overnight security alerts and logs generated by SIEM platforms such as Splunk or IBM QRadar. The analyst triages these alerts, separating false positives from genuine threats, and escalates critical incidents to the incident response team. This continuous monitoring ensures that no threat goes unnoticed, even during off-hours.

Throughout the day, the analyst conducts **vulnerability assessments** using tools like Nessus or Qualys, scanning organizational systems for known weaknesses. When vulnerabilities are discovered, the analyst works with IT teams to prioritize and apply patches. They also perform **threat intelligence analysis**, researching emerging attack techniques, reviewing indicators of compromise (IoCs), and updating detection rules accordingly.

Incident response is another core function. When a security event occurs — whether a phishing attempt, malware infection, or unauthorized access — the analyst follows established playbooks to contain, eradicate, and recover from the incident. Detailed documentation and reporting are maintained for every incident to support forensic analysis and regulatory compliance.

Analysts also contribute to **security awareness training**, educating employees about phishing, social engineering, and safe computing practices. They review and update security policies, conduct access control audits, and participate in tabletop exercises to test the organization's incident response readiness. Collaboration with network engineers, system administrators, and management is essential to align security measures with business objectives.

At the end of the day, the analyst prepares shift handoff notes, documents findings, and updates dashboards and metrics that track the organization's security health. The role requires continuous learning — staying current with certifications (CompTIA Security+, CEH, CISSP), threat landscapes, and evolving attack methodologies.

Key Daily Activities Summary: Alert monitoring and triage, vulnerability scanning, threat intelligence analysis, incident response, security policy review, employee training, access control audits, reporting, and continuous learning.

7. Challenges Faced

Information Overload: Researching five major cyberattacks required sifting through numerous sources with conflicting or outdated information. Cross-referencing multiple reputable sources was essential to ensure accuracy.

CIA Triad Application: Mapping each attack to specific CIA Triad components required careful analysis, as many attacks violate multiple components simultaneously. Distinguishing primary from secondary violations required deeper understanding of attack mechanics.

Network Diagram Design: Creating a comprehensive yet clear network diagram that accurately represents security layers while remaining visually accessible was a balancing exercise between technical detail and readability.

Tool Selection: The cybersecurity tool landscape is vast. Selecting three representative tools that cover different aspects of security (network analysis, scanning, and SIEM) required understanding the broader toolset categories.

8. Conclusion

This Week 01 assignment provided a comprehensive introduction to the core pillars of cybersecurity. By researching real-world cyberattacks, I gained practical insight into how threat actors exploit vulnerabilities and the devastating consequences of security failures. The CIA Triad analysis deepened my understanding of how to evaluate and categorize security incidents by the specific principles they violate.

Exploring cybersecurity tools like Wireshark, Nmap, and Splunk highlighted the importance of having the right instruments for detection, analysis, and response. The secure home network design exercise reinforced the concept of defense-in-depth — applying multiple security layers to minimize risk.

Understanding the daily responsibilities of a cybersecurity analyst clarified the breadth and depth of this career path, from technical incident response to policy development and employee training. This foundational knowledge will serve as the bedrock for the more advanced topics covered in the upcoming weeks of the IT-SIMPLERA Cyber Security Analyst Internship Program.

Key Takeaway: Cybersecurity is not a single solution but a continuous process of layered defense, vigilant monitoring, rapid response, and ongoing education.

End of Report — IT-SIMPLERA Cyber Security Analyst Internship Program — Week 01